WORKED SOLUTION

STEP II 2002, Question 2

Pure • Reciprocal Polynomial Equations

STEP II 2002, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q2

The key idea

Look at the coefficients from the outside in: 1 and 1, then 5 and -5, then 4 in the middle. Coefficients that match or flip sign in pairs mean that dividing by the middle power of z leaves only combinations of $z^k + z^{-k}$ and $z^k - z^{-k}$, and each of these is a polynomial in $w = z - z^{-1}$. That halves the degree. The degree-8 equation has the same structure, and once reduced it turns out to have it *again*, so the trick is used twice.

Solution

The quartic

Since $z = 0$ is not a root (the constant term is 1), we may divide by z^2:

$$z^2 + 5z + 4 - 5z^{-1} + z^{-2} = 0, \qquad \text{that is,} \qquad \left(z^2 + z^{-2}\right) + 5\left(z - z^{-1}\right) + 4 = 0.$$

With $w = z - z^{-1}$ we have $w^2 = z^2 - 2 + z^{-2}$, so $z^2 + z^{-2} = w^2 + 2$. Substituting,

$$(w^2 + 2) + 5w + 4 = 0 \quad \Longleftrightarrow \quad w^2 + 5w + 6 = 0,$$

as required. This factorises as $(w + 2)(w + 3) = 0$, so $w = -2$ or $w = -3$.

To get back to z, multiply $z - z^{-1} = w$ by z:

$$z^2 - wz - 1 = 0.$$

- $w = -2$: $z^2 + 2z - 1 = 0$, so $z = -1 \pm \sqrt{2}$.
- $w = -3$: $z^2 + 3z - 1 = 0$, so $z = \frac{1}{2}\left(-3 \pm \sqrt{13}\right)$.

These four distinct values are the roots of the quartic:

$$z = -1 \pm \sqrt{2}, \qquad z = \frac{-3 \pm \sqrt{13}}{2}.$$

The equation of degree 8

Again $z = 0$ is not a root, so we divide by z^4 and pair the coefficients symmetrically about the middle:

$$2\left(z^4 + z^{-4}\right) - 3\left(z^3 - z^{-3}\right) - 12\left(z^2 + z^{-2}\right) + 12\left(z - z^{-1}\right) + 22 = 0.$$

(Check the pairs: z^8 and z^0 both have 2; z^7 has -3 and z^1 has $+3$; z^6 and z^2 both have -12; z^5 has 12 and z^3 has -12.)

We need each bracket in terms of $w = z - z^{-1}$. We already have $z^2 + z^{-2} = w^2 + 2$. Cubing,

$$w^3 = z^3 - 3z + 3z^{-1} - z^{-3} \quad \Longrightarrow \quad z^3 - z^{-3} = w^3 + 3w,$$

and squaring $z^2 + z^{-2}$,

$$z^4 + z^{-4} = \left(z^2 + z^{-2}\right)^2 - 2 = (w^2 + 2)^2 - 2 = w^4 + 4w^2 + 2.$$

Substituting,

$$\begin{aligned} 0 &= 2(w^4 + 4w^2 + 2) - 3(w^3 + 3w) - 12(w^2 + 2) + 12w + 22 \\ &= 2w^4 - 3w^3 - 4w^2 + 3w + 2. \end{aligned}$$

Ask yourself We are left with a quartic in w. Have we gained anything?

Yes: look at its coefficients, 2, -3, -4, 3, 2. They have exactly the same pattern as before (the outer pair equal, the next pair opposite in sign), so the same trick works one level down. Since $w = 0$ is not a root, divide by w^2:

$$2\left(w^2 + w^{-2}\right) - 3\left(w - w^{-1}\right) - 4 = 0.$$

Put $v = w - w^{-1}$, so that $w^2 + w^{-2} = v^2 + 2$:

$$2(v^2 + 2) - 3v - 4 = 0 \quad \Longleftrightarrow \quad 2v^2 - 3v = 0 \quad \Longleftrightarrow \quad v(2v - 3) = 0.$$

So $v = 0$ or $v = \frac{3}{2}$. Now unwind, using $w^2 - vw - 1 = 0$:

- $v = 0$: $w^2 = 1$, so $w = \pm 1$.
- $v = \frac{3}{2}$: $2w^2 - 3w - 2 = 0$, that is $(2w + 1)(w - 2) = 0$, so $w = 2$ or $w = -\frac{1}{2}$.

Finally unwind once more with $z^2 - wz - 1 = 0$:

- $w = 1$: $z^2 - z - 1 = 0$, so $z = \frac{1}{2}\left(1 \pm \sqrt{5}\right)$.
- $w = -1$: $z^2 + z - 1 = 0$, so $z = \frac{1}{2}\left(-1 \pm \sqrt{5}\right)$.
- $w = 2$: $z^2 - 2z - 1 = 0$, so $z = 1 \pm \sqrt{2}$.
- $w = -\frac{1}{2}$: $2z^2 + z - 2 = 0$, so $z = \frac{1}{4}\left(-1 \pm \sqrt{17}\right)$.

Each quadratic $z^2 - wz - 1 = 0$ has discriminant $w^2 + 4 > 0$, so all these roots are real, and the eight values are distinct. Since a polynomial of degree 8 has at most eight roots, these are all of them:

$$z = \frac{1 \pm \sqrt{5}}{2}, \qquad z = \frac{-1 \pm \sqrt{5}}{2}, \qquad z = 1 \pm \sqrt{2}, \qquad z = \frac{-1 \pm \sqrt{17}}{4}.$$

Checks

- Each pair of roots from $z^2 - wz - 1 = 0$ has product -1, which is a quick check on the arithmetic: for example $(1 + \sqrt{2})(1 - \sqrt{2}) = -1$.
- For the degree-8 equation the sum of the roots should be $\frac{3}{2}$ (minus the z^7 coefficient over the z^8 coefficient). The four pairs contribute $1 - 1 + 2 - \frac{1}{2} = \frac{3}{2}$. Good.
- Numerically, $z = \frac{1}{4}(-1 + \sqrt{17}) \approx 0.7808$ makes the degree-8 polynomial vanish to within 10^{-9}, as do the other seven roots and the four roots of the quartic.

Common slips

- Writing $z^3 - z^{-3} = w^3$: the cross terms give the extra $3w$.
- Stopping at the quartic in w and trying to factorise it by trial, instead of noticing that its coefficients have the same symmetry.
- Giving only the values of w (or v) and forgetting to return to z.

Beyond STEP

Polynomials whose coefficients read the same (or the same up to alternating sign) backwards are called *reciprocal* or *palindromic*; their roots come in pairs related by $z \mapsto 1/z$ or $z \mapsto -1/z$, and the substitution here is exactly the reduction this symmetry allows. In Galois theory this is the observation that the equation is invariant under a group of transformations of the roots, so it can be solved in stages through a tower of smaller equations, which is what happened here: degree 8, then 4, then 2. Allenby's *Rings, Fields and Groups* is a good place to see these ideas develop.

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